

Assignment #4

Student name:

Course: *DATA 407/STAT 507* – Professor: *Dr. Xiaoping Shi (xiaoping.shi@ubc.ca)*

Due date: *April 1, 2026*

Instructions.

For DATA 407, you should submit answers to Questions 1 through 7, and for STAT 507, you should submit answers to all questions, Questions 1 through 8.
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Possible multiple choice.

1. (2 marks) Given a population $Y_1, \dots, Y_N$, the empirical distribution function is $F_N(t) = \frac{1}{N} \sum_{i=1}^N I(Y_i \leq t)$. Take a SRS of size n . The estimate of $F_N(t)$ is $\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n I(Y_i \leq t)$. What is the variance of $\hat{F}_n(t)$?

- a. $\frac{1}{n} \frac{N}{N-1} F_N(t) [1 - F_N(t)]$.
- b. $\frac{1}{n} F_N(t) [1 - F_N(t)]$.
- c. $(1 - \frac{n}{N}) \frac{1}{n} F_N(t) [1 - F_N(t)]$.
- d. $(1 - \frac{n}{N}) \frac{1}{n} \frac{N}{N-1} F_N(t) [1 - F_N(t)]$.

2. (2 marks) Is the histogram an estimate of density?

- a. Yes.
- b. No.

3. (2 marks) Which of the following statements are true about the Jackknife?

- a. Bias reduction.
- b. Variance estimation.
- c. Cross-validation.
- d. Resampling

4. (2 marks) In Bootstrapping, is the resampling done with replacement?

- a. Yes.
- b. No.

5. (2 marks) Let $X_1^*, \dots, X_n^*$ be an iid Bootstrapping sample from $x_1, \dots, x_n$. What's the mean value of $\frac{1}{n} \sum_{i=1}^n X_i^*$ under the above sampling scheme?

- a. $\frac{1}{n-1} \sum_{i=1}^n x_i$.
- b. $\frac{1}{n} \sum_{i=1}^n x_i$.
- c. $\frac{1}{n+1} \sum_{i=1}^n x_i$.
- d. $\frac{1}{n^2} \sum_{i=1}^n x_i$.

6. (2 marks) Let $X_1^*, \dots, X_n^*$ be an iid Bootstrapping sample from $x_1, \dots, x_n$. What's the variance of $\frac{1}{n} \sum_{i=1}^n X_i^*$ under the above sampling scheme?

- a. $\frac{1}{n} \sum_{i=1}^n (x_i - \sum_{j=1}^n x_j / n)^2$.
- b. $\frac{1}{n} \sum_{i=1}^n (x_i - \sum_{j=1}^n x_j / (n-1))^2$.
- c. $\frac{1}{n^2} \sum_{i=1}^n (x_i - \sum_{j=1}^n x_j / (n-1))^2$.
- d. $\frac{1}{n^2} \sum_{i=1}^n (x_i - \sum_{j=1}^n x_j / n)^2$.

7. (3 marks) Suppose that it is desired to estimate $\theta = 1/\sigma^2$ from a series of observations taken from a normal distribution $N(0, \sigma^2)$. Let $t_n = \frac{n-1}{\sum_{i=1}^n (x_i - \bar{x}_n)^2}$. Find the Jackknife estimate t'_n .

8. (3 marks) Let $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$ with $n = 50$, $\mu = 5$ and $\sigma = 1$. Suppose the estimate of cv is $\hat{c}v = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}}{\frac{1}{n} \sum_{i=1}^n X_i}$. Use the Bootstrap method to estimate the variance of $\hat{c}v$.

```
set.seed(1)
n=50;mu=5;sigma=1
x=rnorm(n,mu,sigma)
asy.var=(sigma^4/mu^4+3*sigma^2/(4*mu^2)-sigma^2/(4*mu^2))/n
x.B=NULL
for(b in 1:10000){
#use the Bootstrap method to estimate the variance
#compare the estimated variance to the asymptotic variance asy.var
}
```